

Pattern Matching and Machine Learning for Audio Signal Processing

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Goals of Chapter 3

- Fourier transforms of DT-signals
- Fourier series of periodic signals
- Lossless vs. lossy signal operations
- Fundamental linear systems
- LTI systems
- Convolution theorem
- FIR and IIR filters
- Impulse and frequency response
- Low-pass, high-pass, and band-pass filters
- z -transform

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Chapter 3: Discrete Time Signals and Digital Filters – 2 / 33

Spaces of DT-signals

- In the last chapter we discussed finite signals and the Discrete Fourier Transform. The DFT maps a **time dependent signal** $x \in \mathbb{C}^N$ to its **frequency dependent spectrum** $X = \text{DFT}_N x \in \mathbb{C}^N$. Note that in this case, the signal and the spectral domain coincide, only the interpretation of the parameters differ: time vs. frequency. Moreover, the DFT is invertible, thus the signal x can be recovered from its spectrum X .
- We are going to extend the Fourier analysis to DT-signals $x : \mathbb{Z} \rightarrow \mathbb{C}$. To this end, we introduce suitable signal and spectral spaces. We start with signal spaces.

$$\ell^p(\mathbb{Z}) := \{x : \mathbb{Z} \rightarrow \mathbb{C} \mid \|x\|_p := \left(\sum_{n \in \mathbb{Z}} |x(n)|^p \right)^{1/p} < \infty\}, \quad \text{for } 1 \leq p < \infty,$$

$$\ell^\infty(\mathbb{Z}) := \{x : \mathbb{Z} \rightarrow \mathbb{C} \mid \|x\|_\infty := \sup_{n \in \mathbb{Z}} |x(n)| < \infty\} \quad (\text{bounded DT-signals})$$

- All these spaces are normed linear vector spaces. Moreover, if $1 \leq p < q \leq \infty$, then $\ell^p(\mathbb{Z})$ is a proper subspace of $\ell^q(\mathbb{Z})$. In addition, $\langle x|y \rangle := \sum_{n \in \mathbb{Z}} x(n)\overline{y(n)}$ defines an inner product on $\ell^2(\mathbb{Z})$ and $\|x\|_2 = \sqrt{\langle x|x \rangle}$.

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Fourier transform of DT-signals

- Recall that the spectrum X of $x \in \mathbb{C}^N$ is obtained by evaluating the polynomial $p(z) := \sum_{n=0}^{N-1} x_n z^n$ at all N -th roots of unity: $X = \left(p(\omega^0), p(\omega^1), \dots, p(\omega^{N-1}) \right)^\top$, where $\omega := e^{-2\pi i/N}$.
- The Fourier transform of DT-signals $x : \mathbb{Z} \rightarrow \mathbb{C}$ is the limit $|N| \rightarrow \infty$. Thus in the present case, the Fourier transform is a function of a continuum of frequencies.
- The Fourier transform $X = \hat{x}$ of a DT-signal $x \in \ell^2(\mathbb{Z})$ is defined by

$$X(\omega) := \sum_{n \in \mathbb{Z}} x(n) e^{-2\pi i \omega n}, \quad \text{for } 0 \leq \omega < 1.$$

- X is an element of $L^2([0; 1)) := \{f : [0; 1) \rightarrow \mathbb{C} \mid \int_0^1 |f(\omega)|^2 d\omega < \infty\}$, which can be viewed as the space of all 1-periodic functions of finite 2-norm equipped with the inner product $\langle f|g \rangle = \int_0^1 f(\omega) \overline{g(\omega)} d\omega$.

Theorem 1 (Plancherel). The Fourier transform $x \mapsto X$ is an isomorphism $\ell^2(\mathbb{Z}) \rightarrow L^2([0; 1))$ of inner product spaces. In particular, $\langle x|y \rangle = \langle \hat{x}|\hat{y} \rangle$.

Fourier series

Theorem 2. The family of functions $\left([0; 1) \ni \omega \mapsto e^{2\pi i k \omega} \right)_{k \in \mathbb{Z}}$ forms a complete ON-system in $L^2([0; 1))$. Thus every $f \in L^2([0; 1))$ can be written as a **Fourier series**

$$f(\omega) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{2\pi i k \omega}, \quad (\omega \in [0; 1)),$$

with **Fourier coefficients** $\hat{f}(k) = \int_0^1 f(\omega) e^{-2\pi i k \omega} d\omega$.

Example. We compute the Fourier series of the box function

$$f(\omega) = \begin{cases} 1 & \text{if } \omega \in [0; 0.5) \\ 0 & \text{if } \omega \in (0.5; 1). \end{cases}$$

Obviously $f \in L^2([0; 1))$ and $\hat{f}(0) = \int_0^1 f(\omega) d\omega = 0.5$. For $0 \neq k \in \mathbb{Z}$ we obtain

$$\hat{f}(k) = \int_0^{0.5} e^{-2\pi i k \omega} d\omega = -\frac{1}{2\pi i k} e^{-2\pi i k \omega} \Big|_{\omega=0}^{\omega=0.5} = -\frac{1}{2\pi i k} (1 - (-1)^k).$$

Thus $f(\omega) = 0.5 - \sum_{k \in \mathbb{Z}, \text{odd}} \frac{1}{\pi i k} e^{2\pi i k \omega}$. ◇

Lossless vs. lossy signal operations

- Signal analysis methods like
 - ◆ **Fourier series** and **Fourier-Transforms**yield important information about the original signal, e.g., about the contained harmonic oscillations.
- All of the resulting new representations are **lossless**:
 - ◆ The original signal may be perfectly reconstructed.
- The signal is not changed but only transferred to an alternative representation.
- However, many applications **require to change** the original signal:
 - ◆ **Attenuation or emphasis of spectral characteristics**, e.g., for **noise reduction**,
 - ◆ **Compression, source separation, time stretching**, ...
- In this chapter we introduce fundamental mechanisms for manipulating signals.
 - ◆ Most of these manipulations go along with a **loss** of information.

Linear systems

- In general, every mechanism for modifying a signal is called a **system**.
- Formally, a **system** is specified by a mapping $T : E \rightarrow A$, transforming an input signal $x \in E$ to an output signal $y \in A$. Here, E and A denote suitable signal spaces.

$$x \longrightarrow \boxed{\text{System } T} \longrightarrow y.$$

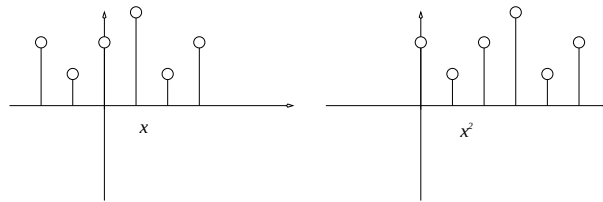
- The simplest, yet very important class of systems are the **linear systems**. Here, the spaces E and A are linear while the mapping $T : E \rightarrow A$ is $\mathbb{C}$ -linear, i.e., for all $x, y \in E$ and all $\lambda \in \mathbb{C}$:

$$T[x + y] = T[x] + T[y] \quad \text{and} \quad T[\lambda x] = \lambda T[x].$$

- We use the mathematical term **operator** interchangeably with the term system.
- **Examples of linear signal spaces:** $\mathbb{C}^N$, $\ell^\infty(\mathbb{Z})$, and $\ell^p(\mathbb{Z})$, for $1 \leq p < \infty$.
- We are going to introduce fundamental linear operators $L : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$. For each L we will also compare the Fourier transforms of $x \in \ell^2(\mathbb{Z})$ and Lx .

Time shift operators

- The **time-shift operator** T^k , $k \in \mathbb{Z}$, defined by $T^k[x](n) = x^k(n) := x(n - k)$ is a linear (lossless) operator $\ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$.



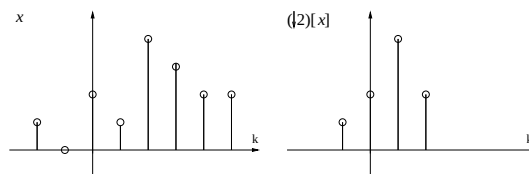
Theorem 3 (Shift Theorem). Let $x \in \ell^2(\mathbb{Z})$ and let $y = x^k$ denote the k -shift of x . Then

$$Y(\omega) = e^{-2\pi i \omega k} \cdot X(\omega).$$

Proof. $Y(\omega) = \sum_{n \in \mathbb{Z}} y(n) e^{-2\pi i \omega n} = \sum_{n \in \mathbb{Z}} x(n - k) e^{-2\pi i \omega (n - k + k)} = e^{-2\pi i \omega k} \cdot X(\omega).$ □

Downsampling operators

- The **downsampling operator** $\downarrow M$, $M \geq 2$, defined by $\downarrow M[x](n) := x(M \cdot n)$ is a (lossy) linear operator $\ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$.
- Downsampling $\downarrow M$ only conserves each M -th value of an input signal:



Theorem 4 (Aliasing Theorem). Let $x \in \ell^2(\mathbb{Z})$ and let $y = \downarrow M[x]$. Then

$$Y(\omega) = M^{-1} \sum_{k=0}^{M-1} X\left(\frac{\omega + k}{M}\right).$$

The terms for $k > 0$ are called **aliasing terms**.

Proof of the aliasing theorem*

$$\begin{aligned}
 \frac{1}{M} \sum_{k=0}^{M-1} X\left(\frac{\omega+k}{M}\right) &= \frac{1}{M} \sum_{k < M} \sum_{n \in \mathbb{Z}} x(n) e^{-2\pi i \frac{\omega+k}{M} n} \\
 &= \sum_{n \in \mathbb{Z}} x(n) \frac{1}{M} \sum_{k < M} e^{-2\pi i \frac{\omega+k}{M} n} \\
 &= \sum_{n \in \mathbb{Z}} x(n) e^{-2\pi i \frac{\omega}{M} n} \underbrace{\frac{1}{M} \sum_{k < M} (e^{-2\pi i (n/M)})^k}_{=: Z_n \stackrel{!}{=} \delta_{M|n}, \text{ see below}} \\
 &= \sum_{n \in \mathbb{Z}} x(Mn) e^{-2\pi i \frac{\omega}{M} (Mn)} \\
 &= \sum_{n \in \mathbb{Z}} y(n) e^{-2\pi i \omega n} = Y(\omega).
 \end{aligned}$$

$Z_n \stackrel{!}{=} \delta_{M|n}$: $q := e^{-2\pi i (n/M)} = 1$ iff M divides n . Thus $Z_n = 1$, if M divides n , otherwise $Z_n = 0$, according to the formula $\sum_{k < M} q^k = (q^M - 1)/(q - 1)$, valid for $q \neq 1$, and the fact that $q^M = 1$. $\square$

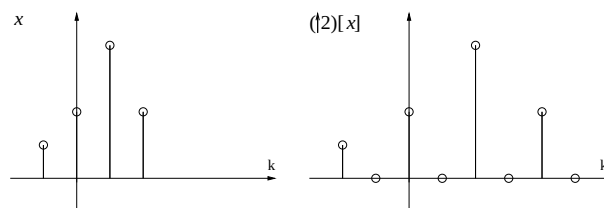
Upsampling operators

- The **upsampling** or **stretching operator** $\uparrow M$, $M \geq 2$, defined by

$$(\uparrow M)[x](n) = \begin{cases} x(n/M), & \text{if } M \text{ divides } n, \\ 0, & \text{otherwise.} \end{cases}$$

is a lossless linear operator $\ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$.

- $\uparrow M$ temporally stretches the input signal x by a factor of M and inserts zeros at all resulting $M - 1$ positions between two original x -values:



Spectral impact of upsampling

Theorem 5 (Stretch Theorem). Let $x \in \ell^2(\mathbb{Z})$, $M \geq 2$, and $y = \uparrow M x$. Then

$$Y(\omega) = X(M\omega).$$

Proof. Note that $y(n) = 0$ unless M divides n . If $n = mM$, then $y(n) = x(m)$. Hence

$$\begin{aligned} Y(\omega) &= \sum_{n \in \mathbb{Z}} y(n) e^{-2\pi i \omega n} = \sum_{m \in \mathbb{Z}} x(m) e^{-2\pi i \omega mM} = \sum_{m \in \mathbb{Z}} x(m) e^{-2\pi i (M\omega)m} \\ &= X(M\omega). \end{aligned}$$

This proves our claim. □

- Thanks to the 1-periodicity of X , the formula $Y(\omega) = X(M\omega)$ says that Y repeats M times an M -fold compressed version of X .

Amplitude modulation

- **Amplitude modulation** with respect to a **carrier signal** $c \in \ell^\infty(\mathbb{Z})$ is the linear operator defined by $m_c[x](n) := c(n) \cdot x(n)$.
- An important special case is the **frequency-shift** E_ω , $\omega \in (0, 1)$, defined by

$$E_\omega[x](n) := e^{-2\pi i \omega n} x(n).$$

- For the frequency signal $f_{\omega_0} = (e^{2\pi i \omega_0 n})_{n \in \mathbb{Z}} \in \ell^\infty(\mathbb{Z})$ one obtains

$$E_\omega[f_{\omega_0}] = f_{\omega_0 - \omega}.$$

- The operators E_ω and $E_{-\omega}$ are inverse to each other with respect to operator composition: $E_{-\omega} \circ E_\omega[x] = E_\omega \circ E_{-\omega}[x] = x$.
- The operator $E_{-\omega}$ is therefore called **demodulation operator**.
- Application: **frequency multiplexing**, e.g., for communication signals.
- ... or to make a bat sound audible to humans!

Spectral impact of modulation

Theorem 6 (Modulation Theorem). Let $x \in \ell^2(\mathbb{Z})$, $\omega_0 \in (0; 1)$, and $y = E_{\omega_0}x$. Then

$$Y(\omega) = X(\omega + \omega_0).$$

(We interpret X as a 1-periodic function.)

Proof. Note that $y(n) = x(n)e^{-2\pi i \omega_0 n}$. Hence

$$\begin{aligned} Y(\omega) &= \sum_{n \in \mathbb{Z}} y(n) e^{-2\pi i \omega n} = \sum_{n \in \mathbb{Z}} x(n) e^{-2\pi i \omega_0 n} e^{-2\pi i \omega n} = \sum_{n \in \mathbb{Z}} x(n) e^{-2\pi i (\omega_0 + \omega) n} \\ &= X(\omega_0 + \omega). \end{aligned}$$

This proves our claim. □

■ Thus a modulation of x with a frequency signal causes a cyclic shift in the spectral domain.

LTI systems and convolution

- Of particular importance are systems that commute with every time shift T^k .
- A system $S: \ell^p(\mathbb{Z}) \rightarrow \ell^r(\mathbb{Z})$ is called **time invariant**, if $S \circ T^k = T^k \circ S$ holds for all $k \in \mathbb{Z}$. This means, for all $k \in \mathbb{Z}$ and all $x \in \ell^p(\mathbb{Z})$: $S[x^k] = S[x]^k$.
- A system that is both linear and time invariant is called an **LTI-system**.
- LTI-systems have an important characterization in terms of convolutions.
- For two signals $x, y: \mathbb{Z} \rightarrow \mathbb{C}$, the **convolution** of x and y at position $n \in \mathbb{Z}$ is defined by

$$(x * y)(n) := \sum_{k \in \mathbb{Z}} x(k) y(n - k) = \sum_{k, \ell \in \mathbb{Z}: k + \ell = n} x(k) y(\ell).$$

- **Note:** $x * y$ only exists if x and y fulfill suitable conditions. Here comes an important example.

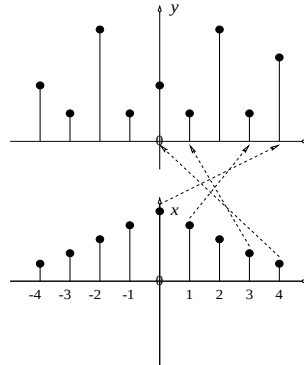
Theorem 7. Let $p \in [1, \infty]$. If $h \in \ell^1(\mathbb{Z})$, $x \in \ell^p(\mathbb{Z})$, then $h * x \in \ell^p$ and $\|h * x\|_p \leq \|h\|_1 \cdot \|x\|_p$.

Illustration of the convolution

Let $y: \mathbb{Z} \rightarrow \mathbb{C}$ denote a signal and let x be a probability density function on $\mathbb{Z}$, i.e. $x: \mathbb{Z} \rightarrow [0, 1]$ and $\sum_{n \in \mathbb{Z}} x(n) = 1$. Then

$$(x * y)(n) = \sum_{k \in \mathbb{Z}} x(k)y(n - k) = \sum_{k, \ell \in \mathbb{Z}: k + \ell = n} x(k)y(\ell)$$

may be interpreted as weighted average of y in a neighborhood of position n . We illustrate this for $n = 4$:



Properties of the convolution

Theorem 8. Let x, y, z denote signals $\mathbb{Z} \rightarrow \mathbb{C}$. Under the assumption that the following convolutions and products exist, the following laws hold:

- **Commutativity:** $x \cdot y = y \cdot x$, $x * y = y * x$,
- **Associativity:** $(x \cdot y) \cdot z = x \cdot (y \cdot z)$, $(x * y) * z = x * (y * z)$,
- **Distributivity:** $(x + y) \cdot z = x \cdot z + y \cdot z$, $(x + y) * z = x * z + y * z$.

From this and Theorem 7 it follows that for a fixed $h \in \ell^1(\mathbb{Z})$

$$C_h[x] := h * x$$

yields a well-defined and linear **convolution operator** $C_h: \ell^p(\mathbb{Z}) \rightarrow \ell^p(\mathbb{Z})$. Moreover, this operator is Lipschitz-continuous, i.e., there exists a constant $c \geq 0$ such that $\|C_h[x] - C_h[x']\|_p \leq c \cdot \|x - x'\|_p$ holds. In our case, $c = \|h\|_1$. (Note that that Lipschitz-continuity implies the classical notion of continuity.)

The convolution theorem

We describe the spectral impact of convolutions.

Theorem 9. Let $h \in \ell^1(\mathbb{Z})$ and $x \in \ell^2(\mathbb{Z})$. Then $y = h * x \in \ell^2(\mathbb{Z})$ and

$$\widehat{h * x} = \widehat{h} \cdot \widehat{x}.$$

Thus the convolution of h and x modifies the spectrum of x point-wise:

$$Y(\omega) = H(\omega) \cdot X(\omega) \quad (\omega \in [0; 1)).$$

Proof.

$$\begin{aligned} \widehat{h * x}(\omega) &= \sum_{n \in \mathbb{Z}} (h * x)(n) e^{-2\pi i \omega n} \\ &= \sum_{n \in \mathbb{Z}} \sum_{k \in \mathbb{Z}} h(k) x(n - k) e^{-2\pi i \omega (n - k + k)} \\ &= \sum_{k \in \mathbb{Z}} h(k) e^{-2\pi i \omega k} \sum_{n \in \mathbb{Z}} x(n - k) e^{-2\pi i \omega (n - k)} \\ &= \widehat{h}(\omega) \cdot \widehat{x}(\omega). \quad \square \end{aligned}$$

Time-shifts are convolution operators

Theorem 10. Let $p \in [1, \infty]$. For $k \in \mathbb{Z}$ the time-shift operator $T^k : \ell^p(\mathbb{Z}) \rightarrow \ell^p(\mathbb{Z})$ coincides with the convolution operator associated with the k -shifted delta-function δ^k :

$$T^k[x] = x^k = x * \delta^k, \quad \text{for all } x \in \ell^p(\mathbb{Z}).$$

Proof. Evaluating the left hand side at $n \in \mathbb{Z}$ yields $T^k[x](n) = x(n - k)$. For the corresponding value of the right hand side we obtain

$$\begin{aligned} (x * \delta^k)(n) &= \sum_{\ell \in \mathbb{Z}} x(\ell) \delta^k(n - \ell) = \sum_{\ell \in \mathbb{Z}} x(\ell) \delta_{k, n - \ell} \\ &= \sum_{\ell \in \mathbb{Z}} x(\ell) \delta_{\ell, n - k} = x(n - k) = T^k[x](n). \end{aligned}$$

Hence we have proved that the time-shift operator T^k and the convolution operator C_{δ^k} coincide on $\ell^p(\mathbb{Z})$. $\square$

Corollary 11. Let $p \in [1, \infty]$. The convolution operator corresponding to $h \in \ell^1(\mathbb{Z})$ is a continuous LTI-system $C_h : \ell^p(\mathbb{Z}) \rightarrow \ell^p(\mathbb{Z})$. Moreover, if C_h is applied to the unit impulse δ , then the system outputs h : $C_h[\delta] = h$. (Therefore h is called the **impulse response**.)

Frequency response

- The spectral characteristics of a convolution operator C_h , $h \in \ell^1(\mathbb{Z})$, are described by the Fourier transform of its impulse response:

Theorem 12. Let $f_\omega := (e^{2\pi i \omega n})_{n \in \mathbb{Z}} \in \ell^\infty(\mathbb{Z})$ denote the frequency signal corresponding to $\omega \in [0, 1)$. If $h \in \ell^1(\mathbb{Z})$, then

$$C_h[f_\omega] = H(\omega) \cdot f_\omega,$$

where $H(\omega) = \hat{h}(\omega) = \sum_{n \in \mathbb{Z}} h(n) e^{-2\pi i \omega n}$. Thus $H(\omega)$ is an eigenvalue of C_h belonging to the eigenvector f_ω . (Therefore, $\hat{h}$ is called the **frequency response** of the system C_h .)

Proof. For all $n \in \mathbb{Z}$ we obtain

$$\begin{aligned} C_h[f_\omega](n) &= \sum_{k \in \mathbb{Z}} h(k) f_\omega(n-k) = \sum_{k \in \mathbb{Z}} h(k) e^{2\pi i \omega (n-k)} \\ &= \sum_{k \in \mathbb{Z}} h(k) e^{-2\pi i \omega k} e^{2\pi i \omega n} = \left(\sum_{k \in \mathbb{Z}} h(k) e^{-2\pi i \omega k} \right) f_\omega(n) = H(\omega) \cdot f_\omega(n). \quad \square \end{aligned}$$

Amplitude response & phase response

- Let $T = C_h$, $h \in \ell^1(\mathbb{Z})$. Writing the frequency response of T in polar coordinates

$$H(\omega) = |H(\omega)| \cdot e^{2\pi i \Phi_h(\omega)} \quad (\omega \in [0; 1)),$$

we introduce the following notions:

- ◆ **amplitude response** $\omega \mapsto |H(\omega)|$,
- ◆ **phase response** $\omega \mapsto \Phi_h(\omega)$,
- ◆ **spectral energy density** $\omega \mapsto |H(\omega)|^2$.

- Note that the phase response is only defined for frequencies ω where $H(\omega) \neq 0$.

Properties of the frequency response

The frequency response H of a filter $h \in \ell^1(\mathbb{Z})$ contains the **spectral information** concerning h and has the following properties:

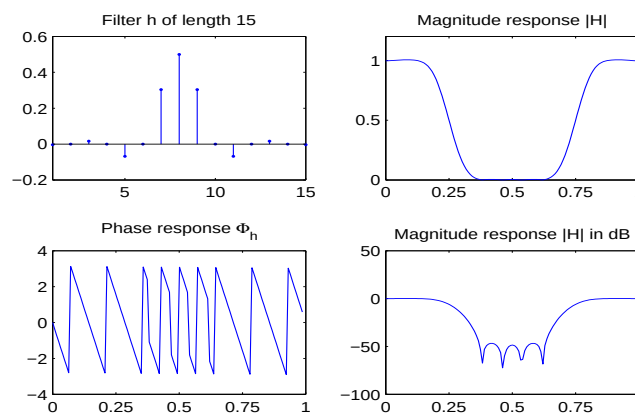
- For real-valued h , $\overline{H(\omega)} = H(-\omega)$. Hence the amplitude response then is an even function, $|H(\omega)| = |H(-\omega)|$, while the phase response is an odd function, $\Phi_h(-\omega) = -\Phi_h(\omega)$.
- The frequency response is 1-periodic. As the frequency response satisfies $\overline{H(\omega)} = H(-\omega)$ in case of real-valued h , in this case all information on H is contained in the restriction of H to the interval $[0, \frac{1}{2}]$, as for $\omega \in [0, \frac{1}{2}]$ one obtains

$$H(1 - \omega) = H(-\omega) = \overline{H(\omega)}.$$

- Spectral information regarding low (high) frequencies correspond to the $\omega \in [0, \frac{1}{2}]$ near 0 (near $\frac{1}{2}$ respectively).

Illustration: amplitude- & phase-response

The following illustration shows the amplitude response $|H|$ and the phase response Φ_h of a filter h of length 15. The filter coefficients $\neq 0$ of h are $h(1) = -0.0036 = h(15)$, $h(3) = 0.0161 = h(13)$, $h(5) = -0.0682 = h(11)$, $h(7) = 0.3038 = h(9)$ and $h(8) = 0.5$. The behavior of $|H|$ is furthermore represented in the decibel-scale.



FIR, IIR & causal filters

- Let $h \in \ell^1(\mathbb{Z})$. Then $h(n)$ is called the **n -th filter coefficient** of the LTI-system $T = C_h$.
- T is called an **FIR-filter** or **FIR-system** (Finite Impulse Response), if only finitely many filter coefficients are non-zero.
- Otherwise, it is called **IIR-filter** or **IIR-system** (Infinite Impulse Response).
- T is called **causal**, if $h(n) = 0$, for all $n < 0$. Causality is important in view of real-time applications (as one cannot access the future values of a signal).
- Usually, T and its impulse response h are identified.
- Causal FIR-filters $T = C_h$ of **order** N and **length** $N + 1$ are filters that can be written as

$$T[x](n) = \sum_{\ell=0}^N h(\ell)x(n-\ell)$$

with filter coefficients $h(0), \dots, h(N)$, $h(N) \neq 0$ and $h(0) \neq 0$. The n -th value of the output signal $T[x]$ then only depends on the *past* values $x(n-1), \dots, x(n-N)$ as well as on the *present* value $x(n)$ of the input signal x . (Hence one speaks of *causality*.)

Ideal low-, high- & band-pass filters

- $h \in \ell^1(\mathbb{Z})$ describes an (ideal) **low-pass filter** with a cut-off frequency $\omega_0 \in [0, \frac{1}{2}]$, if

$$|H(\omega)| \approx \begin{cases} 1 & \text{if } |\omega| \in [0, \omega_0] \\ 0 & \text{if } |\omega| \in (\omega_0, \frac{1}{2}] \end{cases}$$

- Correspondingly, h is called an (ideal) **high-pass filter**, if

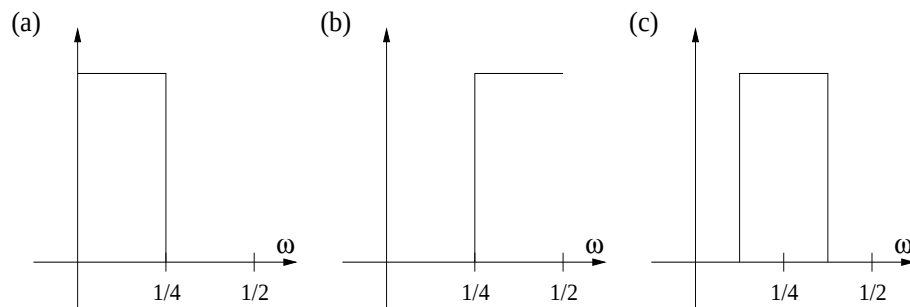
$$|H(\omega)| \approx \begin{cases} 1 & \text{for } |\omega| \in [\omega_0, \frac{1}{2}] \text{ and} \\ 0 & \text{for } |\omega| \in [0, \omega_0). \end{cases}$$

- Finally, h is an (ideal) **band-pass filter** for the cut-off frequencies $0 < \omega_0 < \omega_1 < \frac{1}{2}$, if

$$|H(\omega)| \approx \begin{cases} 1 & \text{for } |\omega| \in [\omega_0, \omega_1] \text{ and} \\ 0 & \text{for } |\omega| \in [0, \omega_0) \cup (\omega_1, \frac{1}{2}]. \end{cases}$$

Illustration of ideal filters

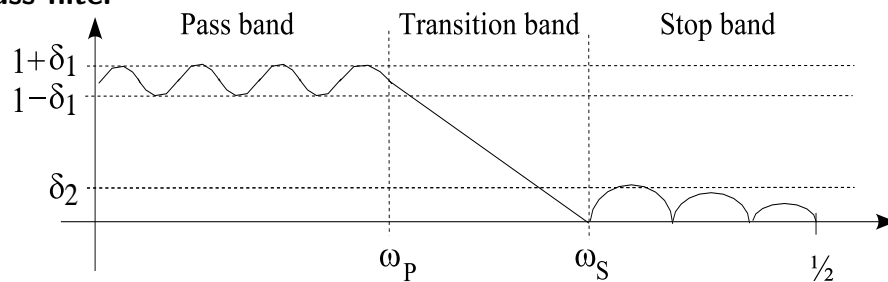
- The following figure shows the amplitude responses of ideal (a) low-pass-, (b) high-pass-, and (c) band-pass filters.



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Real low-pass filter



- The typical task of designing a filter may be specified by the following parameters:

1. maximum magnitude of ripple δ_1 in the pass-band
2. maximum magnitude of ripple δ_2 in the stop-band
3. band-edge frequency ω_P of the pass-band
4. band-edge frequency ω_S of the stop-band

Goal: construct a filter H with a frequency response H satisfying, or closely approximating, these specifications.

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Chapter 3: Discrete Time Signals and Digital Filters – 27 / 33

The z-transform

The values of discrete signals may be interpreted as coefficients of polynomials (finite signals) or formal power series (infinite signals) in some variable, say z .

- For example, $x = (1, 3, 5) \rightsquigarrow 1 \cdot z^1 + 3 \cdot z^2 + 5 \cdot z^3 =: \mathbf{x}(z)$.
- Then, the convolution rather obviously corresponds to a multiplication of polynomials.
- Furthermore, a substitution $z \mapsto e^{2\pi i \omega}$ naturally yields a connection to the Fourier spectrum.
- Such a representation is useful and frequently used in filter design.
- This is formalized by the notion of z -transform or Laurent-series.

Definition of the z-transform

Definition. Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ denote a signal. The **z-transform** $Z[x] =: X$ of x is the function $X : D_x \rightarrow \mathbb{C}$ defined by

$$X(z) := \sum_{n \in \mathbb{Z}} x(n) z^{-n},$$

where D_x denotes the largest open subset of $\mathbb{C}$ such that for all $z \in D_x$ the sum on the right hand side exists.

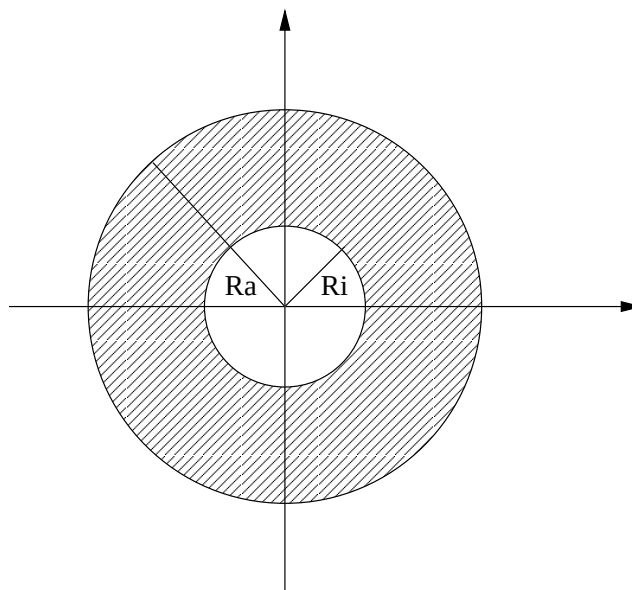
The **region of convergence** (ROC) is disc-shaped. The inner radius is given by

$$R_i(x) = \overline{\lim}_{n \rightarrow \infty} |x(n)|^{1/n},$$

while the outer radius is

$$R_a(x) = \frac{1}{\overline{\lim}_{n \rightarrow \infty} |x(-n)|^{1/n}}.$$

Illustration of region of convergence



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Properties of the z-transform (1)

Theorem 13. Let x, y, h denote signals $\mathbb{Z} \rightarrow \mathbb{C}$ with associated z -transforms X, Y, H . Then the following properties hold:

1. **Linearity:**

$$Z(\lambda x + y) = \lambda Z(x) + Z(y)$$

(on the common region of convergence!)

2. **Delay-property:** Let $y = x^k = T^k[x]$ denote the k -shifted version of x , then

$$Y(z) = z^{-k} X(z).$$

3. **Multiplication by an exponential:** let $0 \neq a \in \mathbb{C}$ and $y(n) = a^n \cdot x(n)$, then $D_y = |a| \cdot D_x$ and

$$Y(z) = X(z/a).$$

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Properties of the z-transform (2)

Theorem 14. Let x, y, h denote signals $\mathbb{Z} \rightarrow \mathbb{C}$ with associated z -transforms X, Y, H . Then the following properties hold:

4. **Conjugation-property:** let $y(n) = \overline{x(n)}$, then $D_y = D_x$ and

$$Y(z) = \overline{X(\overline{z})}.$$

5. **Time-reversal:** Let $y(n) = x(-n)$, then $D_y = D_x^{-1}$ and

$$Y(z) = X(z^{-1}).$$

6. **Convolution theorem:** Let $y = h * x$, then $D_h \cap D_x \subseteq D_y$ and

$$Y(z) = H(z) \cdot X(z).$$

Proof. Exercise!

Transfer function

Definition. The z -transform $H(z) = \sum_{n \in \mathbb{Z}} h(n)z^{-n}$ of the impulse response h of a continuous LTI-system T , $T[x] = h * x$, is called the **transfer-function** of the system.

Remarks.

- For $h \in \ell^1(\mathbb{Z})$, the Fourier transform $\hat{h}$ coincides with the z -transform of h restricted to the unit circle:

$$\hat{h}(\omega) = \sum_{n \in \mathbb{Z}} h(n)e^{-2\pi i \omega n} = H(e^{2\pi i \omega}).$$

- By abuse of notation, we have written above $H(\omega)$ instead of $H(e^{2\pi i \omega})$.